

59) How many different truth tables of compound propositions are there that involve the propositional variables p and q ?

Soln: Let x be a general arbitrary compound proposition, having the propositional variables p, q and the logical operators $\neg, \wedge$ and $\vee$ only. Since, by Ex-47, $\neg, \wedge$ and $\vee$ form a functionally complete collection of logical operators, any compound proposition involving only the propositional variables p, q can be represented in the form of x .

p	q	x
T	T	—
T	F	—
F	T	—
F	F	—

For each combination of assignment of truth values to p, q (there are 4 such combinations as shown in the table), x can either take up the truth value True or False, $\therefore x$ has 2 possible choices of Truth values.

That is, for each row in the table, x has 2 possible choices of Truth values.
 $\therefore$ Total no. of truth tables assignable = $2 \times 2 \times 2 \times 2 = 16$

60) Show that if p, q and r are compound propositions such that p and q are logically equivalent and q and r are logically equivalent, then p and r are logically equivalent.

Soln: Given, $(p \equiv q) \wedge (q \equiv r)$ is True. By defn. of logical equivalence, $(p \leftrightarrow q) \wedge (q \leftrightarrow r)$ is a tautology. By defn. of tautology, the given compound proposition is true irrespective of the truth values of p, q and r .

By defn. of AND, $(p \leftrightarrow q)$ is True and $(q \leftrightarrow r)$ is True
 By defn. of biconditional, ① Both p, q are True OR ② Both p, q are False
 By defn. of biconditional, ② Both q, r are True OR ③ Both q, r are False
 If p, q are True, r is also True. If p, q are False, r is also False.

$\therefore p$ and r are having the same Truth values
 $\therefore p \leftrightarrow r$ is True and $p \equiv r$